

Module 03: Perception in Embodied

Learning Objectives

1. Define **Perception** within the context of Embodied.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between Agents and Cognition.

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In Embodied, we see Perception manifest in: * **Specific Example 1:** When I ride a bicycle and approach a narrow gap between two parked cars, I do not calculate widths -- I perceive the gap directly as "passable" or "too tight" through a felt sense of my body's width in relation to the opening; this is ecological perception through the Markov blanket, where the generative model transforms visual information into motor-relevant affordances scaled to my body's dimensions, generating the perception of navigability rather than raw spatial measurements. * **Specific Example 2:** A rock climber scanning a cliff face perceives handholds and footholds as a connected path of reachable surfaces, feeling in her shoulders and fingers the effort each hold will require before she touches it; this embodied perception is the generative model predicting the proprioceptive and tactile consequences of grasping each surface, so that visual perception of the rock is inseparable from the body's simulation of the movement it affords -- seeing and feeling are unified at the Markov blanket of the moving body.

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.